VGU ROBOTIC PROJECT — USE CASE 08 (UC8)

# ROBOTIC ARM CONVEYOR PICK-AND-PLACE WITH OBSTACLE AVOIDANCE

## MILESTONE 1 DEFENSE & PERFORMANCE VERIFICATION

System Setup: UR10e (6-DOF) + 3.6m Conveyor + 2-Finger Claw + 3 Sorting Bins
Software Stack: Unity 2022.3 LTS (PhysX 4.x) + ROS Noetic (MoveIt) + ROS-TCP-Endpoint

Student Engineering Team — October 2026

PRESENTATION OVERVIEW

# Milestone 1 Presentation Agenda

- 01. Project Cover: Overview, platform stack, and milestone scope
- 02. Presentation Agenda: 15 structured technical evaluation slides
- 03. Baseline & Commitments: Baseline achievements from Week 04
- 04. Milestone 1 Scope: 100% fulfillment of commitments (R1-R4)
- 05. System Architecture: Bidirectional Unity <-> ROS-TCP pipeline
- 06. Feedback & Issue Resolutions: Faculty claw feedback & runtime fixes
- 07. Workcell Re-engineering: Option 1 bilateral layout & clearances
- 08. Kinematics Optimization: 90° wrist lock & pure base SCARA sweep
- 09. PhysX & Multi-Item Flow: Friction coasting & 2-3 item flow
- 10. 7-Step State Machine: Autonomous closed-loop sequencer
- 11. Benchmark QA (70 Trials): 100% grasp/place success rate
- 12. Technical Defenses: Faculty committee engineering Q&A;
- 13. Week 06 Objectives: R5 conveyor robustness & seeded spawner
- 14. Milestone 2 Roadmap: Vision classifier & static obstacle avoidance
- 15. Acceptance Conclusion: Formal Milestone 1 sign-off request

BASELINE FOUNDATION

# Baseline Achieved in Week 04 vs Commitments for Milestone 1

## BASELINE ACHIEVED IN WEEK 04

• ROS-Unity Bridge: Port 10000 TCP connection active via ros_tcp_endpoint.
• Coordinate Conversion: Verified FLU mapping: X_ROS = Z_Unity, Y_ROS = -X_Unity, Z_ROS = Y_Unity.
• Perception Topics: /detected_object_pose & /detected_object_type streaming at 10 Hz.
• MoveIt Inverse Kinematics: Pre-grasp hovering feasibility confirmed in RViz.

## COMMITMENTS SET OUT FOR MILESTONE 1

• Trajectory Execution: Stream real-time /joint_states driving physical UR10e in Unity.
• Grasping Mechanism: Implement attach/parenting logic linking target cube to tool0.
• Task Controller: Construct autonomous closed-loop sequencer connecting perception & motion.
• Multi-Bin Sorting: Route red, yellow, and blue cubes to dedicated destination bins.

SCOPE FULFILLMENT

# Milestone 1 Scope: 100% Commitment Fulfillment Status

## 100% COMMITMENT FULFILLMENT

• Trajectory Execution (R2/R3): 50 Hz streaming to 6 ArticulationBodies with zero sagging (100% DONE).
• Grasping Mechanism (R4): Industrial Attach/Parenting logic in GripperController.cs (100% DONE).
• Closed-Loop Sequencer (R4): 7-step state machine (task_state_machine.py) (100% DONE).
• 3-Bin Sorting (R4): Red (Left), Yellow (Right), Blue (Rear) bins routed reliably (100% DONE).

## HIGH-VALUE WORKCELL ENHANCEMENTS

• Kinematic Optimization: Wrist locked 90° downward & pure base SCARA sweep (65% cycle time cut).
• Zero-Teleport Physics: Natural friction coasting halts cubes smoothly at X ≈ 0.00m (±3mm).
• Continuous Multi-Item Flow: Spawner maintains 2-3 items on conveyor with 38cm spacing.
• Empirical QA Logging: 70 consecutive trials logged with 100% success rate in trials.csv.

SYSTEM ARCHITECTURE

# Bidirectional Real-Time Communication Pipeline

| Layer | Responsibility | Topic / Protocol | Rate & Direction |
|---|---|---|---|
| Unity Perception | Detects cube, halts conveyor, converts frame to ROS FLU | /detected_object_pose, /detected_object_type | 10 Hz (Unity -> ROS) |
| ROS-TCP-Endpoint | Low-latency network bridge brokering ROS socket packets | TCP Port 10000 Socket Streaming | Real-time (Bidirectional) |
| State Machine | Closed-loop task logic sequencing 7 execution states | task_state_machine.py orchestrator | Event-driven (ROS Node) |
| MoveIt Core | Plans collision-free Cartesian paths & joint trajectories | manipulator MoveGroupCommander | MoveIt <-> State Machine |
| Robot Controller | Maps joint positions to 6 UR10e ArticulationBodies | /joint_states (JointTrajectoryPoint) | 50 Hz (ROS -> Unity) |
| Gripper Controller | Parenting attach logic; animates fingers and clamps cube | /ur10e/gripper_command, /gripper_status | Event-driven (ROS <-> Unity) |

FEEDBACK & ISSUE RESOLUTIONS

# Instructor Feedback and Solving Runtime Engineering Challenges

## INSTRUCTOR FEEDBACK

Requirement: Visible Mechanical Claw

• Faculty Feedback: Instructor requested an authentic mechanical 2-finger claw on tool0 rather than an abstract contact point.
• Procedural Claw (VisualClaw): Designed flange adapter (1.2cm), navy chassis (2.0cm), and status LED (Cyan/Emerald).
• Industrial Gripper: Aircraft aluminum arms (4.0cm) with high-friction rubber pads; animated carriage (9.2cm to 6.1cm span).

## RUNTIME ISSUE 1

Challenge: Missed Cubes in Continuous Flow

• Observed Bug: Over extended runs, cubes occasionally slipped past pick station due to missed trigger collisions.
• Root Cause: OnTriggerEnter is fragile under continuous multi-item flow and physics frame discretization.
• Technical Solution: Continuous active scan (FindLeadingCubeOnConveyor()) + safety stop latch at X >= -0.015m (100% reliable).

## RUNTIME ISSUE 2

Challenge: Conveyor Penetration & Off-Axis Twist

• Observed Bug: Claw slightly penetrated belt during descent; upon grasping, cube twisted off-axis by 13.6°.
• Root Cause: Tool orientation was skewed (0.7854 quaternion), and forced position/rotation snapped cube off-axis.
• Technical Solution: Standardized down_orientation = Quaternion(0.7071, 0.7071, 0, 0), elevated fingertips (+2.3cm), and pure parenting.

WORKCELL ENGINEERING

# Layout Optimization: Front Pick Zone and Flanking Bins

| Component | Selected Setup | Engineering Rationale |
|---|---|---|
| Conveyor | 3.6m long, centered directly in front (X = -0.60m) | Stable transport through dexterous workspace; ample gliding run |
| Pick Zone | Front center (X = -0.60m, Y = 0.00m) | Symmetric approach minimizing arm travel distance & inertial moments |
| Red & Yellow Bins | Left (Y = +0.55m) and Right (Y = -0.55m) | Clear transfer paths eliminating collision hazards with conveyor |
| Blue Bin | Behind robot (X = +0.50m, Y = 0.00m) | Separates third target via pure base shoulder pan rotation |
| Support Surface | 4.0m x 2.0m heavy industrial table | Rigid mechanical foundation for rails, bins, and robot fixtures |

KINEMATICS OPTIMIZATION

# Downward Wrist Orientation and Pure Base SCARA Sweep

## MOTION STRATEGY

• Downward Tool Lock: tool0 remains vertical ([0, 0, -1]) throughout all states via locked wrist (q4 = 90.0°).
• Pre-Grasp Waypoint: Arm approaches +16cm above target before vertical descent.
• Pure Vertical Descent: Final approach moves along Cartesian Z, avoiding contact with rails.
• Lift Before Transfer: Cube clears conveyor (+16cm) before base rotation begins.
• Dominant Shoulder Pan: Transfer uses base rotation while arm profile remains rigid.

## OPTIMAL JOINT VALUES (RADIANS)

• Standby Pose: [0.2945, -1.4760, -2.1278, -1.1022, 1.5708, 0.5291]
• Red Bin (Left): [-1.0741, -1.4731, -2.2727, -0.9602, 1.5708, 0.5291]
• Yellow Bin (Right): [1.7077, -1.4731, -2.2727, -0.9602, 1.5708, 0.5291]
• Blue Bin (Rear): [-2.7858, -1.3576, -2.3651, -0.9833, 1.5708, 0.5291]
• Kinematic Invariance: Shoulder lift, elbow, and wrist stay within 0.15 rad across all bins.

PHYSICS & MULTI-ITEM FLOW

# Natural Friction Coasting Deceleration & Multi-Item Flow

## RUNTIME BEHAVIOUR & DECELERATION

• Transport: Cubes move at rock-steady conveyor velocity (0.30 m/s).
• Detection: Pick-zone sensor actively tracks leading SortableObject cube.
• Stop: At X >= -0.015m, conveyor cuts power. Friction halts cube smoothly at X ≈ 0.00m (±3mm).
• Zero Teleportation: Strictly eliminates coordinate snapping or artificial freezing.
• Resume: Conveyor automatically resumes once cube is detached into destination bin.

## MULTI-ITEM FLOW & VISUAL RIG

• Multi-Item Flow: Spawner maintains 2-3 active cubes on belt simultaneously.
• Anti-Stacking Spacing: Strict 38cm spacing (IsSpawnAreaClear()) prevents head collisions.
• Visual Sync: UV texture scrolling synchronized 1:1 with object motion at 0.30 m/s.
• Rig Details: Ribbed rubber belt, dual aluminum rails, chrome rollers, and gearmotor unit.
• Sensor LED: Dual-color indicator switches Green (running) to Red (halted at pick center).

TASK SEQUENCING

# 7-Step Closed-Loop Task State Machine Execution

| Step | State Name | Primary Action | Safety & Validation Guard |
|---|---|---|---|
| 1 | STANDBY | Arm hovers above pick station (Z = 0.42 m) | Elbow held elevated; wrist strictly downward |
| 2 | APPROACH | Rapid Cartesian descent to Pre-grasp Pose (+16cm) | Clears conveyor rails in < 0.5s; wait=True feedback |
| 3 | DESCENT | Vertical Cartesian descent to contact (+6.5cm) | Fingertips float 2.3cm above belt (zero sinking) |
| 4 | GRASP | Publish gripper command (True); latch object | Attach logic latches cube; MoveIt updates PlanningScene |
| 5 | LIFT | Linear Cartesian lift back to safe height (+16cm) | 10cm vertical lift clears conveyor rails without snagging |
| 6 | MOVE_TO_BIN | Single shoulder pan rotation to destination bin | Non-singular trajectory avoids table collisions |
| 7 | RELEASE | Publish gripper command (False); release cube | Parenting detached; gravity drops cube into bin |
| 8 | READY / REARM | Arm returns to Standby pose over pick zone | Resumes conveyor & triggers spawner for next cycle |

EMPIRICAL QA

# Benchmark Evaluation: 70 Consecutive Sorting Trials

| Performance Metric | Requirement | Empirical Measured Result | Status Evaluation |
|---|---|---|---|
| Total benchmark trials | >= 10 cycles | 70 consecutive cycles logged | Exceeds requirement by 700% |
| Grasp success rate | >= 90.0% | 100.0% (70 / 70 trials) | Perfect reliability; zero drops |
| Bin sorting accuracy | >= 90.0% | 100.0% (70 / 70 trials) | 100% correct bin routing |
| Mean cycle time | < 30.0 s | 11.23 s (Min: 10.8s, Max: 12.1s) | 2.7x faster than threshold |
| Conveyor transport speed | 0.2 - 0.5 m/s | 0.30 m/s rock-steady velocity | Optimal industrial pace |
| Coasting stop deviation | <= 20 mm | +- 3 mm repeatable halt | High physical precision |

TECHNICAL DEFENSES

# Answers to Faculty Committee Engineering Defenses (Q&A;)

## QUESTION 1: WHY ARE PRE-GRASP AND LIFT POSES MANDATORY?

Defense: Industrial workspaces strictly divide into Free-space transit and Fine Cartesian approach. Pre-grasp (+16cm) establishes a collision-free holding waypoint above the moving conveyor. Moving diagonally from afar risks collisions with conveyor guide rails or bin sidewalls.

## QUESTION 2: WHY AVOID FIXED SLEEP DELAYS FOR MOTION TIMING?

Defense: Motion duration varies substantially with distance (1.07 rad to Red bin vs 2.78 rad to Blue bin). Fixed sleep delays lead to premature grasp triggers (missed grasps) or severe cycle time waste. MoveIt trajectory execution feedback (wait=True) confirms physical arrival before next step.

## QUESTION 3: HOW WERE RAGDOLL OSCILLATIONS & SAGGING PREVENTED?

Defense: Three concerted measures: (1) Locking base_link with immovable = true; (2) Disabling gravity (useGravity = false) on arm links; (3) Tuning high PhysX drive stiffness (stiffness = 50000, damping = 1000, forceLimit = 10000).

SPRINT BACKLOG

# Week 06 Technical Objectives & Fulfilling Requirement R5

## R5 CONVEYOR ROBUSTNESS

• 5-Minute Continuous Run: Validate conveyor, spawner, and sensor run non-stop without jamming.
• Velocity Modulation: Support dynamic speed adjustment (0.1 to 0.5 m/s) via ROS parameter service.
• Friction Stability: Eliminate micro-slip across diverse cube weights (0.1kg - 0.5kg).

## DETERMINISTIC SPAWNER

• PRNG Seed Reproducibility: Implement seed parameters for RandomObjectSpawner.cs.
• Guaranteed Batches: Allow exact sequence replay (Red -> Blue -> Yellow) for regression testing.
• Spacing Verification: Guarantee 38cm spacing under all belt acceleration profiles.

## FAULT RECOVERY

• Belt Jam Auto-Clear: Automated reverse-and-clear logic if an unexpected stall occurs.
• Perception Rollback: Fail-safe state machine rollback if object detection frame drops.
• Socket Watchdog: Auto-reconnect handling for ROS-TCP-Endpoint in case of network hiccups.

**WEEK 06 DEFINITION OF DONE: Complete fulfillment of Requirement R5 (Conveyor & Spawner Integration). The workcell runs hands-free for > 5 minutes with zero cube miss, zero jam, and deterministic regression logging.**

MILESTONE 2 ROADMAP

# Milestone 2 (Week 10): Static Obstacle Avoidance & Perception

## R5 & R6: CONVEYOR & VISION

• R5 (Continuous Conveyor): Industrial belt running continuous multi-item queues with zero jam.
• R6 (Computer Vision Pipeline): Camera perception module replacing metadata detection.
• Color Classifier: OpenCV HSV segmentation classifying Red, Blue, and Yellow cubes reliably.

## R7: 3D LOCALIZATION

• Camera-Robot Calibration: Hand-eye calibration linking camera pixel coordinates to UR10e base_link.
• Centroid Extraction: Projecting bounding box centroids to 3D Cartesian coordinates.
• Localization Accuracy: Benchmarked measurement error <= 20mm compared to ground truth.

## R8: STATIC OBSTACLES

• Static Wall Barrier: Introducing physical obstacle wall between pick station and sorting bins.
• MoveIt PlanningScene: Real-time obstacle representation in collision matrix.
• Collision-Free Detour: OMPL planner automatically computes curved bypass paths around the wall.

**MILESTONE 2 (WEEK 10) ACCEPTANCE CRITERIA: STOP-AND-PICK MODE**
**Verification of Requirements R5, R6, R7, and R8 in Stop-and-Pick mode (conveyor stops before grasp). Arm successfully navigates around static obstacle barrier with zero collision across >= 20 benchmark trials.**

MILESTONE 1 CONCLUSION

# Summary of Deliverables & Official Acceptance Sign-Off

## REQUIREMENTS MET

- R1 (Scope & Architecture): Approved
- R2 (UR10e MoveIt Planning): Operational
- R3 (ROS-Unity Bridge): Streaming at 50 Hz
- R4 (Closed-Loop Pick & Place): Autonomous

## EMPIRICAL EVIDENCE

- 70 Consecutive Trials Logged
- 100.0% Grasp Success Rate
- 100.0% Bin Placement Rate
- 11.23s Mean Cycle Time (2.7x faster)

## DELIVERABLES PACKAGE

- Git Tag Release: m1-verified
- QA Evidence: Complete trials.csv log
- Launcher: Clean run_workcell.sh script
- Visuals: Multi-camera demo toolbar

**FORMAL MILESTONE 1 ACCEPTANCE REQUEST: All commitments made for Milestone 1 have been 100% fulfilled and reinforced with faculty feedback solutions. The UC8 sorting workcell is fully verified and ready for formal Milestone 1 sign-off and progression to Week 06 / Milestone 2.**